

Exploring Emotion Expression Through Silent Speech Interface in Public VR/MR: Effects of Automation on User Experience

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Abstract

Silent speech input offers a secure and private alternative to voice input in public spaces. As HMDs become increasingly mobile, the need for integrating silent speech recognition grows to enable seamless interaction in public settings. However, the potential of silent speech-based emotion recognition in VR/MR environments remains underexplored, despite the importance of emotional expression in social interactions. This paper evaluates how different levels of automation in emotional expression—Manual, Semi-auto, and Automatic—affect user experience in VR environments. The results revealed that Semi-auto methods generally offer better efficiency, user control, and socially acceptable interactions.

CCS Concepts

• Human-centered computing → Virtual reality; Usability testing.

Keywords

silent speech, social acceptability, virtual reality, public space, communication, emotion

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1 Introduction

Speech recognition technology has significantly improved voice input on mobile devices, enabling rapid and convenient hands-free access to information. However, its use in public settings still faces

several challenges. Users often hesitate to vocalize commands or engage in voice chats in public settings due to privacy and security concerns, as well as embarrassment when speaking aloud in front of others [9–11, 24]. Also, the performance of speech recognition systems degrades significantly in acoustically complex environments characterized by ambient noise, overlapping conversations, and diverse environmental sounds [19]. Furthermore, current systems struggle to accommodate users with speech impairments, as speech recognition interfaces are predominantly optimized for typical speech patterns and standard pronunciation. These challenges have spurred the development of silent speech recognition technology, which converts non-acoustic features into text without requiring audible sound production, thereby enabling users to engage in text-based communication. This technology captures subtle movements of the lip, tongue, or facial muscles using video-based [1, 4, 7, 21] or sensor-based [20, 23, 25, 29] inputs. Thus, users can communicate solely through lip movements without a need for vocalization while interacting with devices such as mobile phone cameras [27] or physiological sensors [16]. This enables seamless interaction in noise-sensitive or voice-restricted environments. Research has demonstrated that silent speech not only addresses privacy and security concerns but also proves more socially acceptable in public settings [14, 22, 26].

Concurrently, Virtual Reality (VR) and Mixed Reality (MR) head-mounted displays (HMDs) are becoming increasingly mobile, transforming from specialized equipment into everyday tools [2, 5, 12, 13]. As devices like Apple Vision Pro enable users to engage in various activities in public spaces, from attending Zoom meetings in cafes to YouTube live streaming while traveling, the integration of silent speech recognition becomes particularly important [29]. However, a critical challenge emerges at the intersection of silent speech recognition and VR/MR applications: the expression and recognition of emotions, which play an essential role in social interactions [3, 18]. While traditional speech-to-text communication has made significant progress in emotion recognition through vocal cues, studies focusing on silent speech contexts with VR/MR environments are extremely limited. This highlights a critical need for developing emotion-aware systems specifically tailored for silent speech in VR/MR environments.

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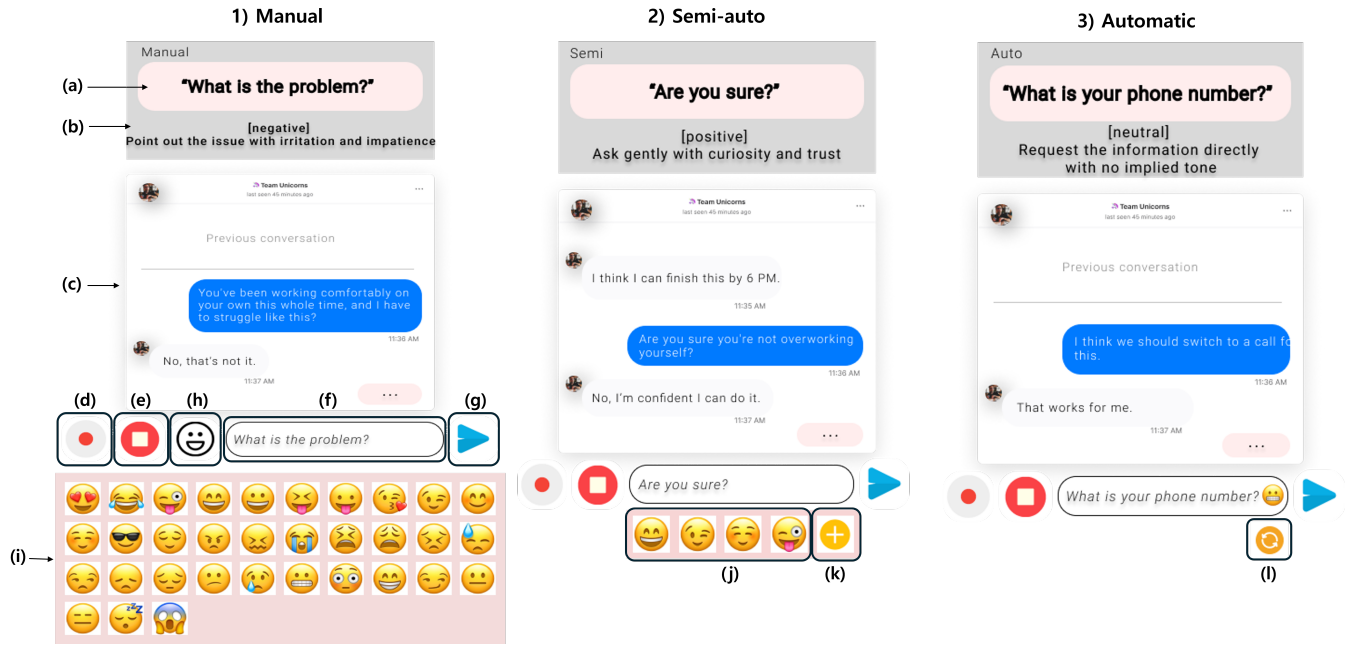


Figure 1: VR User Interface Overview for Each Condition: General UI - (a) phrase, (b) guideline with emotion, (c) dialogue context, (d) record button, (e) stop button, (f) input field, and (g) send button; Manual-specific UI - (h) emoji button and (i) entire emoji panel; Semi-auto-specific UI - (j) candidate emoji panel and (k) plus button; Automatic-specific UI - (l) refresh button.

Among various challenges to address this, our research focuses on a key question: how do users perceive and utilize emotion recognition and expression systems when using silent speech in VR/MR settings? In this work, we specifically focus on three critical factors: efficiency, effectiveness, and social acceptability, which are deeply intertwined. For example, the introduction of automation can improve efficiency and social acceptability by reducing manual input, but it potentially compromises user control and emotional accuracy, ultimately leading to lower effectiveness. This trade-off becomes particularly important in public settings where social attention and privacy concerns influence user behavior. This highlights the importance of balancing efficiency and effectiveness to ensure socially acceptable interactions, with an understanding of how these dimensions interact being crucial for developing systems that users will actually adopt in their daily VR/MR interactions.

To explore this balance, we developed three approaches using emojis as emotional indicators, each offering different levels of automation: Manual (users need to manually select emojis), Semi-auto (the system automatically recognizes emotion and suggests top-4 candidate emojis for user selection), and Automatic (the system automatically recognizes emotion and selects a single emoji for user confirmation). Through within-subject experiments, we explored how varying degrees of control and automation levels affect user experience in public VR scenarios, and identified the most effective approach for emotional expression in terms of balancing social acceptability, effectiveness, and efficiency. Specifically, we focused on identifying an approach that facilitates seamless emotional expression while balancing the level of automation and user control, evaluating user preferences for each method. Additionally,

we conducted a comparative study of traditional voice-based methods to understand the distinct behavioral patterns between silent and vocal communication approaches. In summary, our study investigates emotion expression through a silent speech interface in public VR/MR environments, focusing two key research questions:

- RQ1: How do different levels of automation affect the overall user experience?
- RQ2: How do user experiences differ between silent-speech and voice-speech groups?

2 User Study

In this study, we conducted a comparative evaluation based on the type of speech input and the level of automation in emoji selection. The study was IRB-approved and conducted in accordance with guidelines, ensuring informed consent, anonymity, and data security of the participants throughout the research process.

2.1 Participants

We recruited 24 participants (10 males and 14 females; $M = 25.54$, $SD = 3.52$ years) from the local university, all of whom were right-handed, majoring in industrial engineering, data science, etc. Among them, 19 participants had prior experience with VR devices: 3 with 11+ times, 2 with every 1-2 years, 6 with 3-6 times, and 8 with 1-2 times. Additionally, 19 participants reported prior experience with voice recognition technologies, such as using voice commands to send messages or interacting with systems (e.g., iOS Siri, Amazon Alexa). In contrast, none of the participants had prior experience using silent speech technologies; their familiarity with silent speech

technology was minimal, with an average knowledge score of 1.71 ($SD=0.99$) on a 5-point scale (1: strongly disagree, 5: strongly agree). Participants reported a moderate level of awareness and concern regarding privacy exposure, security risks, and self-consciousness about others' attention when using devices (e.g., phones, VR/MR) in public spaces (user perspective: $M = 3.37$, $SD = 1.05$). Similarly, they reported a moderate level of concern and attentiveness when observing others using such devices in public spaces (observer perspective: $M = 3.21$, $SD = 1.05$).

2.2 Study Design

The recruited 24 participants were divided into two groups of 12 based on the type of speech input interface (between-subject design): *i) Silent-speech*, and *ii) Voice-speech*. Within each group, all participants experienced all the three different conditions (within-subject design): *i) Manual*, *ii) Semi-auto*, and *iii) Automatic*. For experiments, we developed a VR Messenger Application for PICO 4 Enterprise device using Unity 2021.3.13f1 version; see Figure 1 for UI.

Participants completed a task with sending messages using emotional expression through either vocal or silent speech. Each trial began with participants reviewing three elements: a phrase to communicate (Figure 1(a)), guidelines for emotional expression (Figure 1(b)), and the dialogue context (Figure 1(c)). They then recorded their speech by pressing a button (recording: Figure 1(d), stopping: Figure 1(e)), either speaking aloud or using silent speech depending on their assigned condition. After recording, participants verified the accuracy of the speech-to-text transcription (as shown in Figure 1(f)). Following transcription, they engaged with the emoji selection process, which varied based on their assigned condition (Manual, Semi-auto, or Automatic). Once satisfied with both the transcription and emotional expression through emoji, participants completed the interaction by sending their message (Figure 1(g)). For specific UI in the Manual condition, pressing the emoji button (Figure 1(h)) opens the entire emoji panel (Figure 1(i)), similar to the typical process on a mobile phone. In the Semi-auto condition, the candidate emoji panel (Figure 1(j)) appears after recording, and selecting the plus button (Figure 1(k)) opens the entire emoji panel (Figure 1(i)), just as in the Manual condition. Meanwhile, in the Automatic condition, both the transcribed text and a suggested emoji are displayed simultaneously in the input field (Figure 1(f)) after recording, with the emoji being changed upon pressing the refresh button (Figure 1(l)). For a detailed description of the tasks for each condition, see Appendix A.

We selected 36 phrases from the Enron mobile email dataset [28], dividing them equally into three sentiment categories: positive, negative, and neutral. The phrases were carefully chosen to clearly represent each emotion, ensuring that the intended emotional tone was easily recognizable. Each participant sent 12 messages per condition, comprising four phrases from each sentiment category in randomized order. Details on the sentiment phrase sets are in Appendix B. The order of conditions was counterbalanced using a Latin Square design to minimize potential order effects.

In system implementation, (silent) speech and emotion recognition typically rely on visual or sensor data, which require devices such as camera attached to HMD to capture mouth or built-in facial

tracking sensors. Since this paper focuses on the impact of automation levels on user experience, we used a Wizard-of-Oz (WoZ) approach to simulate both (silent) speech and emotion recognition processes. Following prior research on error tolerance in (silent) speech recognition technology [22], we incorporated a controlled word error rate (WER) of 5% for the voice-speech recognition and 15% for the silent-speech recognition, uniformly applied across all experimental phrases. This approach allowed us to focus on the impacts of interaction design on user experience by systematically introducing recognition errors that reflect real-world speech recognition variability within an acceptable margin of error. We employed the emoji set composed of 33 emojis (13 positive, 12 negative, 8 neutral) [15] (see Figure 4). In the Semi-auto and Automatic conditions, the system's emoji selection was predetermined based on emotional categories: the Semi-auto condition presented four randomly selected candidate emojis matching the identified emotion, while the Automatic condition inserted a single emoji for user confirmation. For more details on the system design, please refer to the following video: <https://youtu.be/vtp73dlcDZk>.

2.3 Measurements

We employed well-established questionnaires to assess participants' experiences and perceptions across conditions. We utilized the User Experience Questionnaire (UEQ) [17] for overall usability and the NASA-TLX [6] for workload assessment. For *Perceived Ease of Use* and *Perceived Usability*, we used the Technology Acceptance Model (TAM) questionnaire [8] on a 7-point scale. In addition, we devised a custom questionnaire (custom-Q), evaluating *Social Acceptability*, *Effectiveness*, *Efficiency*, *Willingness to Use* on a 5-point scale (1: strongly disagree, 5: strongly agree). We also measured *Preference Rank* for each condition. Finally, we measured *Task Completion Time (TCT)* for each participant to send a message. For each phrase, the timer started when they pressed the record button (Figure 1(d)) and stopped when they pressed the send button (Figure 1(g)). We then analyzed the average time in seconds for each condition.

2.4 Procedure

The user study took place at campus cafeteria. After greeting the participants, we collected informed consent and demographic information, and explained the experimental tasks in detail. Participants were instructed to speak or silently speak the given phrases with emotion, following the provided guidelines. After confirming their phrases were transcribed into text using (silent) speech recognition, they performed the emoji selection task using the controller under three conditions: Manual, Semi-auto, and Automatic. Participants first completed a training session with different phrase sets from the main experiment across all conditions. During the main experiment, they responded to the questionnaires after completing each condition, and took a 5-minute break between conditions. Upon completing the entire experiment, participants ranked their preferences for the three conditions and participated in a follow-up interview. The entire session per participant lasted approximately 50 minutes, and they were compensated with 8 USD for their time.

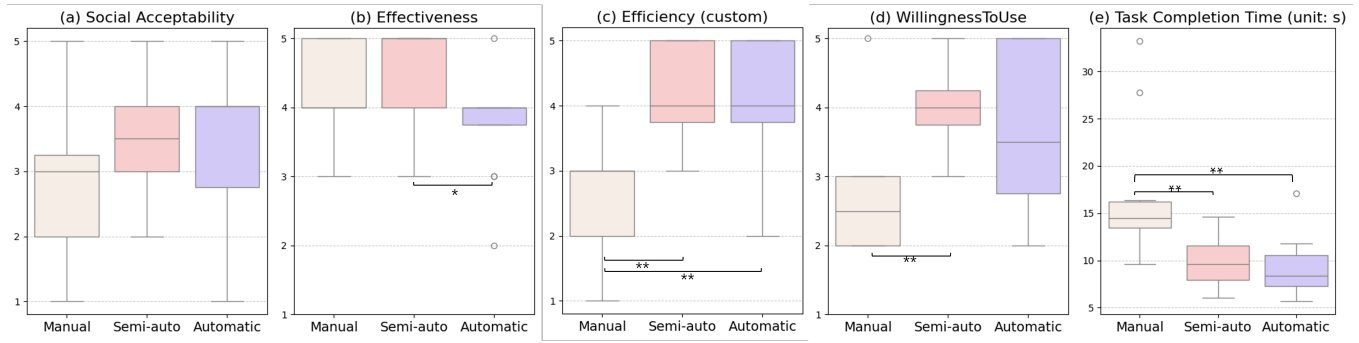


Figure 2: Results within Silent-speech Group (* = $p < .05$, ** = $p < .01$, and *** = $p < .001$).

3 Results

3.1 RQ1: How do different levels of automation and user control affect the overall user experience?

To address RQ1, we analyzed the quantitative results from the silent-speech group across all measures to identify differences between the various control levels in user experience. For the normality test, we conducted the Shapiro-Wilk test for each measure. We used RM-ANOVA for differences across three conditions and Tukey's HSD for post-hoc comparison, if no violations were found. Otherwise, we employed the Friedman test with Bonferroni-adjusted Dunn's tests as post-hoc. The means and standard deviations for all conditions are provided in Table 1 of Appendix C. For results within silent-speech group not shown in the text, please refer to Appendix D.

In our custom-Q, significant differences were observed in *Effectiveness* ($\chi^2(2) = 9.8125, p < 0.01$), *Efficiency* ($\chi^2(2) = 13.6364, p < 0.01$), and *Willingness to Use* ($\chi^2(2) = 9.6923, p < 0.01$), as shown in Figure 2(a-d). In *Effectiveness*, post-hoc tests revealed that Semi-auto ($M = 4.5833$) outperformed Automatic ($M = 3.75$). For *Efficiency*, both Semi-auto ($M = 4.1667$) and Automatic ($M = 4$) outperformed Manual ($M = 2.5833$). For *Willingness to Use*, participants expressed significantly higher intention to use Semi-auto ($M = 4$) than Manual ($M = 2.6667$). In terms of TCT, a significant difference was observed ($\chi^2(2) = 11.5319, p < 0.01$), as shown in Figure 2(e). Participants completed the task significantly faster in both Automatic ($M = 9.2575$) and Semi-auto ($M = 9.9533$) compared to Manual ($M = 16.61$). In UEQ, as shown in Figure 5, there were significant differences in the following metrics: *Attractiveness* ($F(2, 9) = 6.9172, p < 0.01$), *Efficiency* ($F(2, 9) = 8.5713, p < 0.001$), *Stimulation* ($F(2, 9) = 6.5589, p < 0.01$), and *Novelty* ($F(2, 9) = 8.9091, p < 0.01$). Post-hoc comparisons showed that both Semi-auto and Automatic outperformed Manual ($M = 0.0625$) in *Novelty*. As shown in Figure 6, in NASA-TLX, there were significant differences in *Temporal Demand* ($F(2, 9) = 9.3296, p < 0.01$), *Physical Demand* ($\chi^2(2) = 6.7407, p < 0.05$), and *Effort* ($\chi^2(2) = 8.2286, p < 0.05$). Post-hoc tests only showed that participants experienced significantly higher temporal workload in Manual ($M = 33.75$) than Semi-auto ($M = 15$). TAM analysis revealed a significant difference in *Perceived Usability* ($F(2, 9) = 14.4161, p < 0.001$), with post-hoc analysis showing that Semi-auto ($M = 5.75$) was perceived as significantly more usable than Manual ($M = 4.5417$). Regarding *Preference*

Rank, the Semi-auto condition was most frequently rated as the top preference ($N=8$, avg.rank=1.33), followed by the Automatic condition ($N=4$, avg.rank=1.83) and Manual ($N=0$, avg.rank=2.83). Conversely, the Manual condition was deemed the least favorable ($N=10$).

Participants generally perceived the Manual condition as less favorable, with significantly lower in *Novelty*, *Efficiency*, and higher TCT compared to the Semi-auto and Automatic conditions. The lack of automation in Manual condition required users to browse through the entire emoji set and carefully deliberate choices, resulting in increased time consumption and reduced social acceptability. The Manual condition lacked novelty due to its conventional approach. While some participants appreciated the freedom it offered, the overall experience was perceived as inefficient and impractical, with P20 commenting, "The manual mode felt cumbersome, requiring time to view the entire emoji panel and select an emoji, so I would likely avoid using it". In contrast, the Semi-auto and Automatic conditions addressed these limitations by integrating automation to improve efficiency. Participants appreciated the systems' ability to reduce decision-making time by automatically suggesting emoji(s) aligned with their emotional state. However, contrary to expectations, the workload was not significantly reduced in Automatic condition compared to Manual condition. Interestingly, the Semi-auto condition demonstrated the lowest overall workload across all conditions, notably outperforming the Manual condition. Follow-up interviews revealed that while most participants were generally satisfied with the initial suggestion in Automatic condition, some expressed frustration when it did not match their preferences. This misalignment prompted repeated trials to find suitable alternatives, ultimately reducing the system's overall effectiveness. For example, P12 noted, "The suggested emojis were inserted quickly compared to manual mode, allowing me to complete tasks faster and feel less self-conscious in public spaces.", while P6 noted, "The automatic mode suggests a suitable emoji, but often it was not always matching my preferences, requiring extra effort to find an alternative".

In response to RQ1, the Semi-auto method was the most preferred, as participants valued its balance between user control and time efficiency. By offering a set of four suggested emojis, this method significantly streamlines the process compared to the Manual condition, eliminating the need for time-consuming navigation through the full emoji panel. It also retains user agency by allowing participants to make the final selection themselves, providing

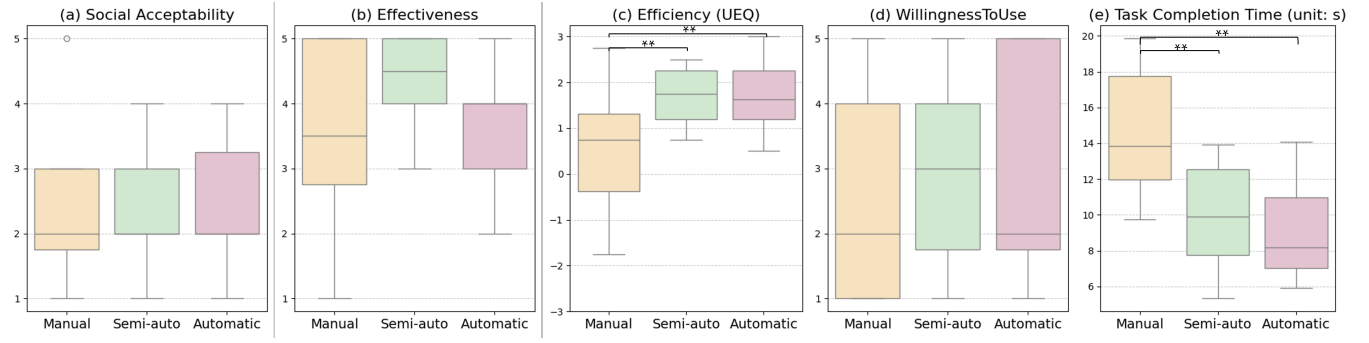


Figure 3: Results within Voice-speech group (* = $p < .05$, ** = $p < .01$, and *** = $p < .001$).

greater control than the Automatic condition. As a result, the combination of enhanced efficiency and preserved user control makes the Semi-auto condition the most favored option, delivering a positive and well-rounded user experience. During the experiment, most participants successfully found their desired emoji among the four options provided without needing a full emoji panel, and many also highlighted the convenience of the plus button. As P2 noted, “In the Semi-auto mode, selecting from four suggested emojis was efficient, and the option to view the full list when needed was also helpful”.

3.2 RQ2: How do user experiences differ between Silent-speech and voice-speech groups?

To address RQ2, we first analyzed how participants within the voice-speech group perceived the three conditions, and examined the overall differences between the silent- and voice-speech groups. For the group comparison, we conducted the Shapiro-Wilk test for the normality test and Levene’s test to assess the homogeneity of variance. If both normality and homogeneity of variance assumptions were met, an independent samples t-test was used; otherwise, Mann-Whitney U tests were conducted. The means and standard deviations per measure for all conditions are provided in Table 2 of Appendix C. For results between voice-speech and silent-speech group, please refer to Appendix E.

In the voice-speech group, similar patterns to the silent-speech group were observed in *TCT* and *Novelty*, with significant differences were also found in UEQ for *Attractiveness* ($F(2, 9) = 10.3058$, $p < 0.001$), *Efficiency* ($F(2, 9) = 12.532$, $p < 0.001$), and *Stimulation* ($F(2, 9) = 4.9965$, $p < 0.05$). Post-hoc tests showed that both Semi-auto ($M = 1.6458$) and Automatic ($M = 1.6667$) outperformed Manual ($M = 0.5417$) in *Efficiency* (Figure 3(c)). While *Efficiency* in custom-Q also showed significant differences ($\chi^2(2) = 10.3333$, $p < 0.01$), but no significant post-hoc results. NASA-TLX results showed significant differences in *Physical Demand* and *Effort*, although no significant post-hoc differences were found, consistent with the silent-speech group. However, unlike the silent-speech group, there was a significant difference in *Mental Demand* ($\chi^2(2) = 8$, $p < 0.05$), though no significant post-hoc results were found. Finally, TAM results showed a significant difference in *Perceived Usability* ($\chi^2(2) = 10.619$, $p < 0.01$), but no significant post-hoc results. In addition, the overall group-level comparison between

the voice-speech and silent-speech groups revealed significant differences. In NASA-TLX, the silent-speech group reported a significantly lower workload compared to the voice-speech group (see in Figure 7). Significant differences were observed in *Mental Demand* ($U = 850$, $p < 0.05$), *Effort* ($U = 858.5$, $p < 0.05$), and *Frustration* ($U = 934.5$, $p < 0.01$). In addition, we found that participants using silent-speech experienced higher levels of social acceptability than those using voice-speech ($U = 381.5$, $p < 0.01$), as shown in Figure 8.

To answer RQ2, both the voice and silent-speech interfaces showed similar patterns, but there were also notable differences in user experience. In the voice-speech interface, similar to the silent speech discussed earlier, the Manual condition performed the worst in *Efficiency*, *TCT*, and *Novelty*. Regarding *Preference Rank*, the Semi-auto was the most preferred condition ($N=8$, avg.rank=1.41), as with the silent-speech interface, followed by Automatic ($N=3$, avg.rank=1.91) and Manual ($N=1$, avg.rank=2.66). Surprisingly, Contrary to expectations, the voice-speech interface showed slightly lower *Effectiveness* compared to the silent-speech interface, though not statistically significant. Unlike the silent condition, the enhanced user control from Manual and Semi-Auto did not result in a significant advantage. This suggests that in a public setting, the need to vocalize emotions in the voice interface may have caused self-consciousness about others hearing their voice, leading to a sense of discomfort. This discomfort likely impaired their ability to effectively engage with the system. In fact, comparing the silent and voice-speech interface, participants reported an overall increased workload when using the voice input, with significantly increased *Mental Demand* score, reflecting the added psychological strain. This increased burden is further reflected in the significantly lower *Social Acceptability* score for the voice-speech interface. As a result, although no significant difference was found, the *Perceived Usability* of the system was lower in the voice-speech interface compared to the silent-speech, leading to a reduced *Willingness to Use* the system.

Based on these results, the silent-speech input facilitated less demanding and more socially acceptable interactions compared to the voice-speech interface. As P18 noted, “In noisy environments, crowded places, or public transportation, silent speech would be more practical than voice input as it wouldn’t disturb others.”

4 Discussion

4.1 Balancing Social Acceptability, Effectiveness, and Efficiency for Public VR Communication

In terms of messaging, follow-up interviews revealed a striking contrast between participants' everyday smartphone habits and their preferences in VR. Only 2 of 24 participants regularly used automated text or emoji recommendation systems on smartphones, the majority (N=14) did not use such features at all. Despite this clear preference for manual input in smartphone interactions, preferences shifted in public VR settings, where participants preferred (semi) automated methods over manual input. The shift in user behavior can be attributed to the unique challenges posed by VR interactions in public settings, including increased physical effort for manual input and heightened social awareness, and the need for improving social acceptability. Semi-automated and fully automated recommendation systems effectively address these VR-specific challenges by minimizing physical movements, improving efficiency through reduced input time, and enhancing effectiveness in conveying emotions. This demonstrates how context-specific demands in VR can fundamentally reshape user needs, prioritizing both social acceptability and interaction efficiency without compromising the effectiveness of emotional expression.

During follow-up interviews, each group was presented with the alternative input method and asked about their preference if both options were available. The results also revealed a strong preference for silent input in public settings from both groups. Most participants (N=16) preferred silent mode, with 9 from the silent-speech and 7 from the voice-speech group, citing three primary concerns: unwanted social attention (N=9), minimizing disturbance to others (N=4), and privacy preservation (N=2). Silent mode was preferred in noisy, crowded, or silent-required environments and for tasks involving sensitive information, as it improved social acceptability and privacy. As P18 noted, *"In noisy/crowded environments like public transport, silent speech technology would be more useful than voice, as it minimizes disruption while maintaining communication"*. In contrast, six participants preferred voice input, with five noting that deliberate lip movements without vocalization might appear more conspicuous than speaking, especially with a VR HMD. They believed that voice input felt more intentional and socially acceptable due to its clear audio feedback. These findings highlight the complex interplay between social acceptability, privacy, and interaction performance in public VR use.

In conclusion, our findings suggest that when using silent-speech input in public settings with HMDs, Semi-auto condition is the most effective for conveying emotions. This combination balances user effort and social acceptability, making it ideal for communication in shared or sensitive environments. By reducing workload, improving effectiveness in emotional expression, and increasing social acceptability, this approach provides a practical solution for enhancing interaction in these contexts.

4.2 Limitations & Future works

This study provides valuable insights into emoji-based emotion expression in public VR environments using silent speech input.

However, there were several limitations. The limited emoji selection made it difficult for participants to accurately express emotions, especially neutral ones. This highlights the need for a broader, contextually relevant emoji set and alternative methods for conveying subtle or ambiguous emotions. Second, our experiment was conducted in a controlled setting where (silent) speech and emotion recognition were simulated using the WoZ method with predefined phrases, and participants were seated. However, real-world applications are likely to involve more complex and dynamic contexts. For instance, in VR conferencing or gaming, conversational contexts would be more varied and less predictable. In addition, users would also likely be moving rather than seated. In such environments, factors like background noise could reduce the accuracy of speech and emotion recognition, which in turn could impact the system's usability and overall user experience. Therefore, future work should prioritize integrating a fully functional system optimized for real-time performance and stability, and explore the feasibility and applicability of our approach in more diverse use cases and real-world scenarios. In addition, this fully integrated system could facilitate future research aimed at supporting more seamless remote communication for users with speech or facial impairments by leveraging XR-based facial reconstruction technology and avatar-based emotion recognition. Lastly, this study focused on public VR environments, leaving the potential of MR and pass-through environments underexplored. These settings, which blend virtual and physical worlds, pose new challenges for emotion expression and interaction, presenting greater social acceptability issues. Future research should explore the integration of silent-speech input with emotion recognition in these contexts.

5 Conclusion

In this paper, we explored emotion expression through silent speech input within public VR environments, focusing on the balance between automation and user control. The study explored three conditions divided by levels of automation in emotion expression: Manual, Semi-auto, and Automatic. The results revealed that the Semi-auto condition, which optimally balanced automation and user control, significantly enhanced effectiveness and efficiency of emotional expression, showing as the most preferred option. Future work needs to address the limitations of this study and develop a fully functional system optimized for real-time performance.

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A The Description of the Tasks for each Condition

- Manual: Manually browse the entire emoji panel that appears when the emoji button is pressed, find, select, and insert an emoji that best represents your emotion.
- Semi-auto: The system will automatically analyze the sentiment of your utterance and suggest four candidate emojis. Select the emoji that you feel best represents your emotion. If none of the suggested emojis are suitable, you can press the plus-button to choose one from the entire emoji panel.
- Automatic: The system will automatically analyze the sentiment of your utterance and add an emoji it considers most appropriate. If the added emoji is not suitable, you can change it to another emoji within that emotion range by pressing the refresh button.

B Emoji & Phrase sets

Positive	
Negative	
Neutral	

Figure 4: Emoji sets for positive/negative/neutral emotions [15]

- Neutral
 - * 9) Is that OK?
 - * 10) How are you?
 - * 11) How do you need it?
 - * 12) I am still here.
- Phrase sets of Automatic condition
 - Positive
 - * 1) This looks fine.
 - * 2) You are the greatest.
 - * 3) This is good I think.
 - * 4) That would be great.
 - Negative
 - * 5) What is this?
 - * 6) Why do you ask?
 - * 7) No can do.
 - * 8) We are all fragile.
 - Neutral
 - * 9) Are you available?
 - * 10) What is your phone number?
 - * 11) I am on my way.
 - * 12) What do you think?

- Phrase sets of Manual condition
 - Positive
 - * 1) Thanks for your concern.
 - * 2) I think that is the right answer.
 - * 3) Good for you.
 - * 4) Can you help me here?
 - Negative
 - * 5) Keep me posted.
 - * 6) I am almost speechless.
 - * 7) Did that happen?
 - * 8) What is the problem?
 - Neutral
 - * 9) What do you hear?
 - * 10) Where are you?
 - * 11) I am trying again.
 - * 12) What is the plan?
- phrase sets of Semi-auto condition
 - Positive
 - * 1) Let it rip.
 - * 2) Are you feeling better?
 - * 3) Are you sure?
 - * 4) See you soon!
 - Negative
 - * 5) What is the mood?
 - * 6) Is it over?
 - * 7) What is the purpose of this?
 - * 8) No surprise there.

C Results of Statistical tests

Table 1: Results of statistical tests within the Silent/Voice group (* = $p < .05$, ** = $p < .01$, and * = $p < .001$).**

Measure: item / Dimension	Silent						Voice					
	Manual		Semi-auto		Automatic		Manual		Semi-auto		Automatic	
	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD
UEQ: Attractiveness	1.125	1.0424	1.9433	0.6166	1.7917	0.8046	0.8608	1.1854	1.5842	0.5296	1.7367	0.8696
UEQ: Perspicuity	1.8958	0.7721	2.1042	0.7108	1.8125	0.8536	1.7708	0.9442	2	0.5436	2.3125	0.5126
UEQ: Efficiency	0.9375	0.7318	1.6458	0.7572	1.5	0.8919	0.5417	1.247	1.6458	0.6258	1.6667	0.7098
UEQ: Dependability	0.5417	0.8177	1.2292	0.5979	0.9792	0.882	0.7708	0.7794	0.8725	0.5796	0.8125	0.7081
UEQ: Stimulation	0.9583	1.2785	1.8125	0.6836	1.7292	0.8289	0.7917	1.1621	1.4375	0.6495	1.4792	0.801
UEQ: Novelty	0.0625	1.3698	1.2292	0.8948	1.2292	0.9321	0	1.4262	1.125	0.5887	1.3125	0.7472
TAM: Perceived Usability	4.5417	1.1831	5.75	0.8087	5.2083	1.1418	3.4306	1.7342	5.0833	1.626	4.875	1.4584
TAM: Perceived Ease of Use	6.0694	0.6608	6.3611	0.6546	6.0833	0.7961	5.7917	0.8169	6.0972	0.7087	6.1389	0.778
NASA-TLX: Mental Demand	20.4167	20.052	10.8333	11.645	15	17.0561	39.1667	28.2709	28.3333	24.1523	19.5833	22.1008
NASA-TLX: Physical Demand	20.4167	18.8846	11.6667	14.1956	15.4167	15.8771	31.6667	26.6572	22.5	25.0907	17.0833	19.2423
NASA-TLX: Temporal Demand	33.75	20.6843	15	11.0782	21.25	18.1064	48.3333	26.1406	25.4167	20.2774	28.3333	16.967
NASA-TLX: Performance	32.0833	20.7209	20.8333	17.1667	27.5	17.5162	36.25	28.5343	37.0833	29.2682	35	20.6706
NASA-TLX: Effort	27.0833	17.7685	12.5	12.7029	19.5833	15.1445	45.8333	27.0381	25.4167	20.9391	28.75	18.9647
NASA-TLX: Frustration	15.4167	13.0486	8.75	11.3067	15.8333	17.5594	40.8333	25.658	26.6667	23.7729	23.75	21.1193
NASA-TLX: Overall	24.8611	15.6462	13.2639	11.4617	19.0972	15.3184	40.3472	22.6732	27.5694	19.6029	25.4167	14.9768
Social Acceptability	2.8333	1.1146	3.5	0.7977	3.3333	1.1547	2.3333	1.1547	2.4167	0.9962	2.5	1.0871
Effectiveness	4.3333	0.6513	4.5833	0.6686	3.75	0.7538	3.5833	1.4434	4.3333	0.7785	3.5	0.9045
Efficiency	2.5833	0.793	4.1667	0.8348	4	0.9535	2.5	1.3143	3.9167	0.793	3.9167	1.0836
Willingness to Use	2.6667	0.8876	4	0.7385	3.5833	1.2401	2.5833	1.5643	2.9167	1.505	2.9167	1.7299
Task Completion Time	16.61	6.89	9.9533	2.711	9.2575	3.0836	14.4708	3.5028	9.9042	2.9154	9.1858	2.8312

Table 2: Results of statistical tests between the Silent-speech and Voice-speech group (* = $p < .05$, ** = $p < .01$, and * = $p < .001$).**

Measure: item / Dimension	Overall				Manual				Semi-auto				Automatic			
	Silent		Voice		Silent		Voice		Silent		Voice		Silent		Voice	
	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD
UEQ: Attractiveness	1.62	0.8913	1.3939	0.9579	1.125	1.0424	0.8608	1.1854	1.9433	0.6166	1.5842	0.5296	1.7917	0.8046	1.7367	0.8696
UEQ: Perspicuity	1.9375	0.7685	2.0278	0.7116	1.8958	0.7721	1.7708	0.9442	2.1042	0.7108	2	0.5436	1.8125	0.8536	2.3125	0.5126
UEQ: Efficiency	1.3611	0.8333	1.2847	1.0267	0.9375	0.7318	0.5417	1.247	1.6458	0.7572	1.6458	0.6258	1.5	0.8919	1.6667	0.7098
UEQ: Dependability	0.9167	0.8062	0.8186	0.6752	0.5417	0.8177	0.7708	0.7794	1.2292	0.5979	0.8725	0.5796	0.9792	0.882	0.8125	0.7081
UEQ: Stimulation	1.5	1.0142	1.2361	0.9276	0.9583	1.2785	0.7917	1.1621	1.8125	0.6836	1.4375	0.6495	1.7292	0.8289	1.4792	0.801
UEQ: Novelty	0.8403	1.194	0.8125	1.1266	0.0625	1.3698	0	1.4262	1.2292	0.8948	1.125	0.5887	1.2292	0.9321	1.3125	0.7472
TAM: Perceived Usability	5.1667	1.143	4.463	1.7321	4.5417	1.1831	3.4306	1.7342	5.75	0.8087	5.0833	1.626	5.2083	1.1418	4.875	1.4584
TAM: Perceived Ease of Use	6.1713	0.6998	6.0093	0.7632	6.0694	0.6608	5.7917	0.8169	6.3611	0.6546	6.0972	0.7087	6.0833	0.7961	6.1389	0.778
NASA-TLX: Mental Demand	15.4167	16.6208	29.0278	25.574	20.4167	20.052	39.1667	28.2709	10.8333	11.645	28.3333	24.1523	15	17.0561	19.5833	22.1008
NASA-TLX: Physical Demand	15.8333	16.3663	23.75	23.9754	20.4167	18.8846	31.6667	26.6572	11.6667	14.1956	22.5	25.0907	15.4167	15.8771	17.0833	19.2423
NASA-TLX: Temporal Demand	23.3333	18.4003	34.0178	23.2631	33.75	20.6843	48.3333	26.1406	15	11.0782	25.4167	20.2774	21.25	18.1064	28.3333	16.967
NASA-TLX: Performance	26.8056	18.5993	36.1111	25.6936	32.0833	20.7209	36.25	28.5343	20.8333	17.1667	37.0833	29.2682	27.5	17.5162	35	20.6706
NASA-TLX: Effort	19.7222	16.0777	33.3333	23.7246	27.0833	17.7685	45.8333	27.0381	12.5	12.7029	25.4167	20.9391	19.5833	15.1445	28.75	18.9647
NASA-TLX: Frustration	13.3333	14.1926	30.4167	24.1239	15.4167	13.0486	40.8333	25.658	8.75	11.3067	26.6667	23.7729	15.8333	17.5594	23.75	21.1193
NASA-TLX: Overall	19.0741	14.6639	31.1111	19.9374	24.8611	15.6462	40.3472	22.6732	13.2639	11.4617	27.5694	19.6029	19.0972	15.3184	25.4167	14.9768
Social Acceptability	3.2222	1.045	2.4167	1.0522	2.8333	1.1146	2.3333	1.1547	3.5	0.7977	2.4167	0.9962	3.3333	1.1547	2.5	1.0871
Effectiveness	4.2222	0.7601	3.8056	1.1166	4.3333	0.6513	3.5833	1.4434	4.5833	0.6686	4.3333	0.7785	3.75	0.7538	3.5	0.9045
Efficiency	3.5833	1.1052	3.4444	1.2533	2.5833	0.793	2.5	1.3143	4.1667	0.8348	3.9167	0.793	4	0.9535	3.9167	1.0836
Willingness to Use	3.4167	1.1052	2.8056	1.5642	2.6667	0.8876	2.5833	1.5643	4	0.7385	2.9167	1.505	3.5833	1.2401	2.9167	1.7299
Task Completion Time	11.9433	5.6181	11.1869	3.8316	16.61	6.89	14.4708	3.5028	9.9533	2.711	9.9042	2.9154	9.2575	3.0836	9.1858	2.8312

D Results for Measures within Silent-speech group

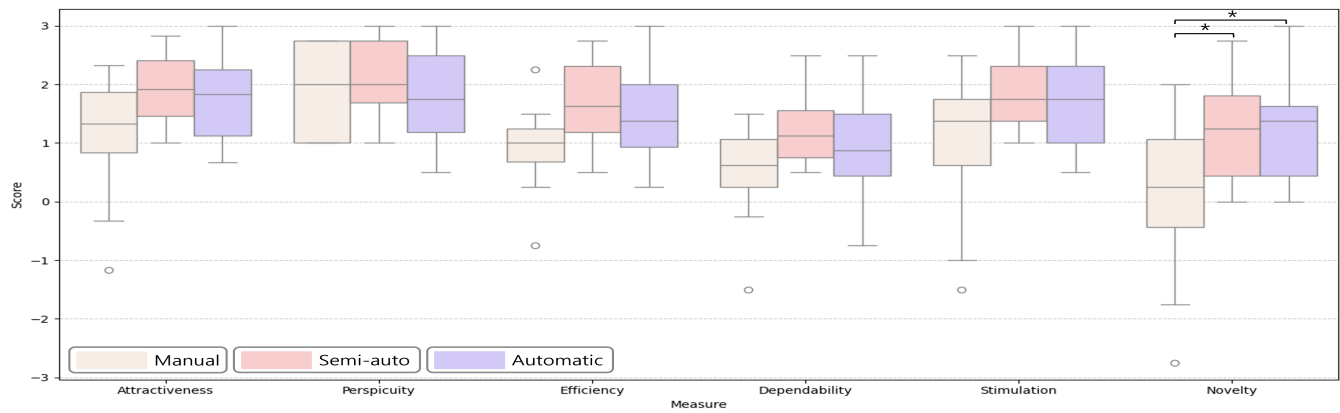


Figure 5: Results of User Experience Questionnaire within Silent-speech group (* = $p < .05$, ** = $p < .01$, and *** = $p < .001$).

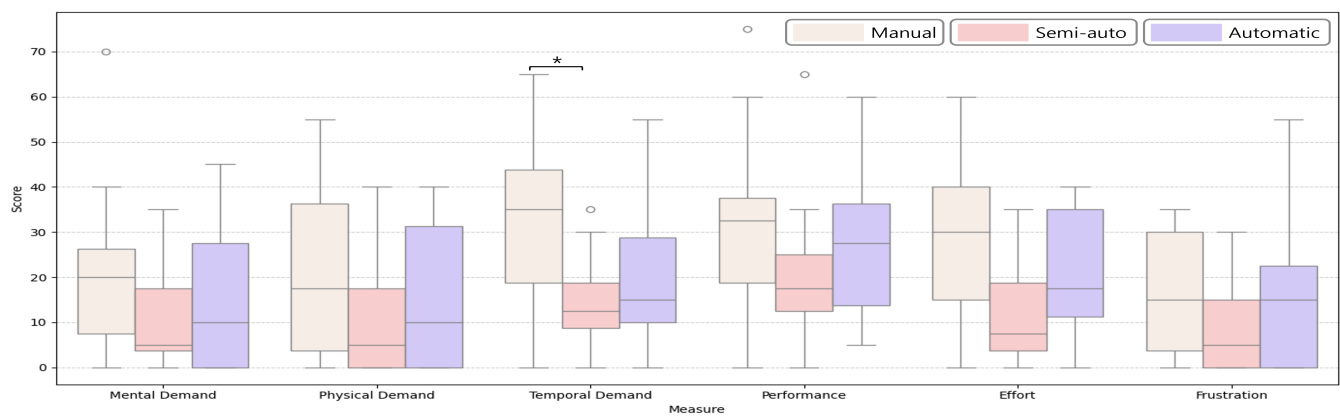


Figure 6: Results of NASA-TLX within Silent-speech group (* = $p < .05$, ** = $p < .01$, and *** = $p < .001$).

E Results for Measures between Voice-speech and Silent-speech group

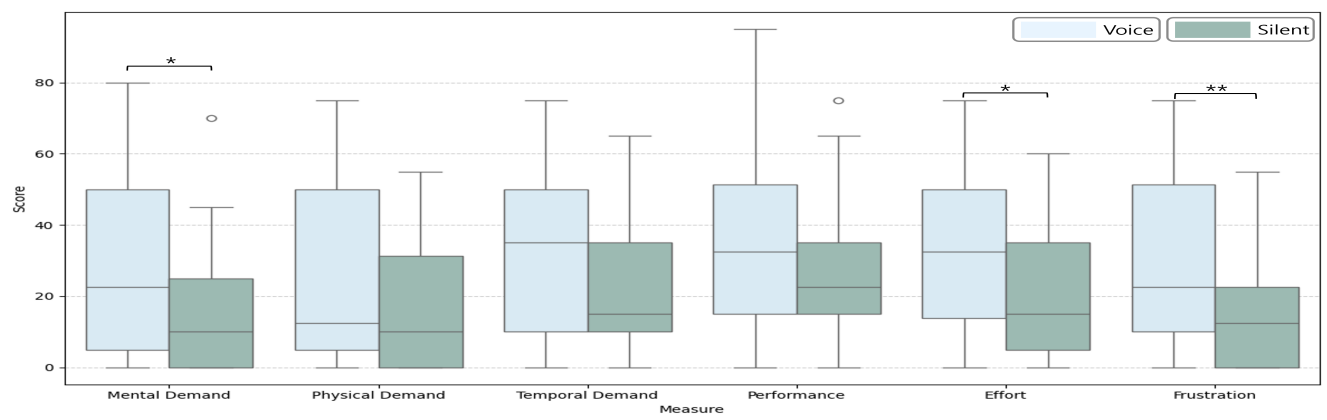


Figure 7: Results of NASA-TLX between Voice-speech and Silent-speech group in Group-level Analysis (* = $p < .05$, ** = $p < .01$, and *** = $p < .001$).

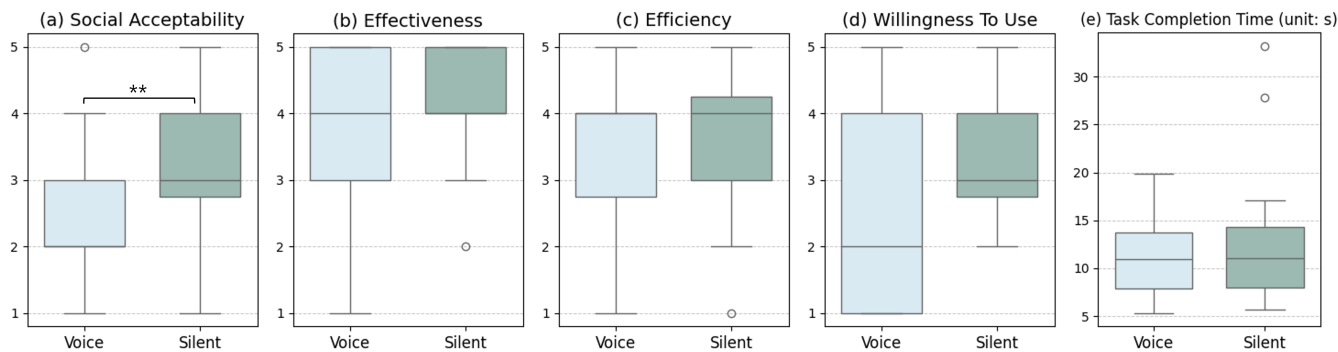


Figure 8: Results of *Social Acceptability*, *Effectiveness*, *Efficiency*, *Willingness To Use*, and *Task Completion Time* between Voice-speech and Silent-speech group in Group-level Analysis (* = $p < .05$, ** = $p < .01$, and *** = $p < .001$).